# BASIC CIVIL & MECHANICAL ENGINEERING

Subject Code: 23A01201T | UNIT – 1 : BASICS OF CIVIL ENGINEERING

## 1. Role of Civil Engineers in Society

> **Civil Engineering** is the oldest branch of engineering – the art and science of **planning, designing, constructing, operating and maintaining** the built and natural environment: buildings, roads, bridges, dams, canals, water supply, sewers, airports, ports, railways, tunnels.

- Provides the **basic infrastructure** needed for human life – shelter, safe water, sanitation, transport, power projects, irrigation.
- Contributes to **economic development** – good roads, ports and railways reduce cost of goods and create jobs.
- Ensures **safety** of structures against load, earthquake, wind, flood & fire.
- Protects the **environment** – water/ waste-water treatment, solid waste management, pollution control.
- Plans **smart cities**, housing for all, disaster-resistant & sustainable construction.
- Famous Indian civil engineer: **Sir M. Visvesvaraya** (Krishnaraja Sagara dam) – **Engineers' Day = 15 September.**

### ❖Responsibilities of a civil engineer

- Survey & investigation → planning → design → estimating & costing → construction supervision → quality control → maintenance.
- Works with architects, contractors, mechanical, electrical engineers & government.

> Civil engineer = builder of the nation's infrastructure

## 2. Various Disciplines of Civil Engineering

- Structural Engineering
- Geo-technical Engineering
- Transportation Engineering
- Hydraulics & Water Resources Engineering

- Environmental Engineering
- (Also: Construction Management, Surveying/ Geomatics, Coastal & Earthquake Engineering)

### ❖(i) Structural Engineering

- Deals with **analysis and design of structures** so that they safely resist loads (dead, live, wind, earthquake, snow).
- **Scope:** RCC and steel buildings, bridges, towers, chimneys, industrial sheds, water tanks, dams, high-rise & tall structures, offshore platforms, earthquake-resistant & prestressed structures.
- Uses: structural analysis software (STAAD, ETABS), IS codes (IS 456 for concrete, IS 800 for steel, IS 1893 for earthquake).
- Concerned with **strength, stability, serviceability, economy.**

### ❖(ii) Geo-technical Engineering

- Study of **soil and rock** behaviour and their use as foundation or construction material.
- **Scope:** soil investigation (boring, SPT), **bearing capacity**, design of **foundations** (shallow – footing, raft; deep – piles, wells), **retaining walls**, slope stability, embankments, earth dams, ground improvement, tunnelling, landslide control.
- Soil properties: grain size, plasticity (Atterberg limits), permeability, shear strength, compaction, consolidation.

### ❖(iii) Transportation Engineering

- Planning, design, construction & operation of **highways, railways, airports, harbours, tunnels, pipelines.**
- **Scope:** traffic engineering, pavement design, highway geometric design, road safety, urban transport (metro, BRTS), railway track design, airport layout, port planning.
- Reduces travel time, cost, accidents and pollution.

### ❖(iv) Hydraulics & Water Resources Engineering

- Study of **water flow, storage, distribution and control.**
- **Scope:** hydrology, **dams & reservoirs, canals, irrigation**, hydropower, flood control, river training, groundwater (wells), water supply schemes, drainage, pumps & turbines, ports/ coastal protection.

- Basic principles: continuity, Bernoulli's equation, flow in pipes & open channels.

### ❖(v) Environmental Engineering

- Protection of environment & public health.
- Scope: water supply & treatment, **waste-water (sewage) treatment**, solid-waste management, air & noise pollution control, **environmental impact assessment (EIA)**, hazardous & e-waste, climate change, green buildings, sustainable development.

## 3. Building Construction & Planning

### ❖Components of a building

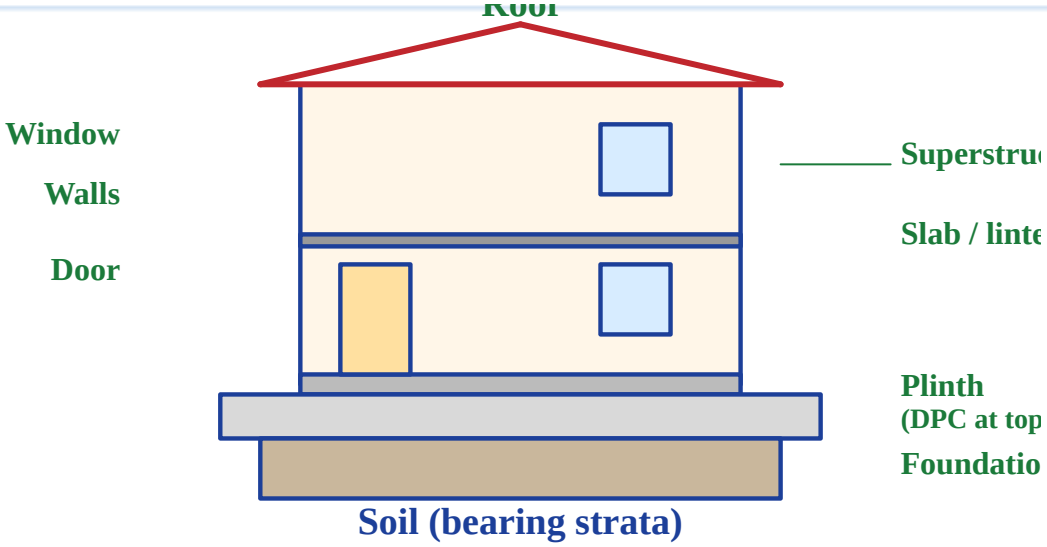



Fig 1.1 Components of a building: sub-structure (below plinth) and super-structure (above)

- **Sub-structure (below ground/ plinth): foundation** – transfers load safely to soil; **plinth** – raised portion above ground level (≥ 450 mm) with **DPC (damp-proof course)** to stop capillary moisture.
- **Super-structure:** walls (load bearing/ partition), columns, beams, slabs, lintels, sills, doors, windows, staircases, roof, floors, finishing (plaster, paint).
- **Foundation types:** isolated footing, combined footing, strip, **raft**, pile, well.
- **Walls** are built with brick/ stone/ concrete blocks in **bonds (English, Flemish)** to avoid vertical continuous joints.
- **Lintel:** beam over openings (doors/ windows). **Sill:** bottom of a window. **Chajja (sunshade).**

### ❖Types of buildings (by use)

- Residential, commercial, educational, institutional (hospital), industrial, assembly (cinema, auditorium), storage.

### ❖Types of construction

- **Load-bearing structure** – walls carry loads (up to 3 floors). **Framed (RCC) structure** – columns & beams carry loads; walls are only partitions (high rise).
- **Composite** – steel + concrete.

## 4. Building Planning

> **Planning** = arranging various rooms/ spaces of a building in a suitable manner to satisfy the needs of the occupant, economically, as per rules (building bye-laws).

### ❖Principles of planning

- **Aspect:** orientation of rooms to get sun, air (bedroom – east/ south-east; kitchen – south-east; living – north/ east; toilet – north-west).
- **Prospect:** pleasing outside view. **Privacy** – from neighbours & within house.
- **Grouping:** related rooms close – kitchen near dining; bedroom with attached toilet.
- **Roominess:** better space utilisation; correct size of rooms.
- **Furniture requirement, sanitation, circulation** (corridors, staircases), **lighting & ventilation** (window area ≥ 1/10 floor area), **flexibility, economy, elegance, practicability.**
- Follow **local bye-laws:** FAR/ FSI, **setbacks**, plot coverage, height limit.

FSI (FAR) = Total built-up floor area of all floors / Plot area

- **Minimum room sizes (NBC):** habitable room ≥ 9.5 $m^2$ (min width 2.4 m); kitchen ≥ 5 $m^2$; bathroom ≥ 1.8 $m^2$; ceiling height ≥ 2.75 m.
- **Drawings:** plan, elevation, section, site plan, foundation plan; scale 1:100.
- **Vaastu / Green building** and **rainwater harvesting**, solar water heater, STP are now included.

# 5. Construction Materials

- Materials used: stone, brick, cement, sand, aggregate, timber, concrete, steel, glass, bitumen, paints, tiles, plastics, composites.
- Selection: strength, durability, cost, availability, workability, appearance, thermal & fire resistance.

## ❖5.1 Cement

> Cement is a fine, grey, binding material that sets and hardens when mixed with water and binds aggregates together. Invented by Joseph Aspdin (1824) – "Portland cement".

- Raw materials: limestone ($CaCO_3$, calcareous) + clay/ shale (argillaceous – silica, alumina) + gypsum (3–5%) added at the end.
- Manufacture (wet or dry process): crush → grind → mix (slurry) → burn in rotary kiln at 1400–1500 °C → clinker → cool → grind with gypsum → bag (50 kg).

## ❖Main compounds (Bogue)

| Compound | Formula | % | Role |
|---|---|---|---|
| Tricalcium silicate ($C_3S$) | $3CaO·SiO_2$ | 40–50 | Early strength, heat |
| Dicalcium silicate ($C_2S$) | $2CaO·SiO_2$ | 25–30 | Later strength |
| Tricalcium aluminate ($C_3A$) | $3CaO·Al_2O_3$ | 5–10 | Flash set; gypsum controls it |
| Tetra-calcium alumino-ferrite ($C_4AF$) | $4CaO·Al_2O_3Fe_2O_3$ | 10 | Colour |

## ❖Types of cement (BIS)

- OPC (Ordinary Portland Cement) 33, 43, 53 grade – (28-day compressive strength 33/43/53 $N/mm^2$); IS 269 / 8112 / 12269.
- PPC (Portland Pozzolana Cement) – 15–35% fly ash; low heat, good in marine, cheaper – IS 1489.
- PSC (Slag cement), Rapid hardening, Low heat, Sulphate-resisting, White cement, Hydrophobic, High alumina.

### ❖Properties & tests

- Fineness (≥ 225 $m^2$/kg, sieve test), Consistency (Vicat – 26–33% water), Setting time: initial ≥ 30 min, final ≤ 600 min (10 h), Soundness (Le-Chatelier – expansion ≤ 10 mm), Compressive strength of 70.6 mm cement mortar cubes (3, 7, 28 days), specific gravity 3.15.
- Uses: mortar, plaster, concrete, grouting, masonry, flooring.
- Store in dry place, off the ground, away from walls, in stacks ≤ 10 bags; use within 3 months.

### ❖5.2 Aggregates

> Inert granular materials (sand, gravel, crushed stone) which form 60–75% of concrete volume.

- Fine aggregate: passes 4.75 mm IS sieve, retained on 75 µm (sand; river sand, M-sand, crushed sand). Zones I–IV.
- Coarse aggregate: retained on 4.75 mm sieve – gravel, crushed granite/ basalt; sizes 10, 12.5, 20, 40 mm (20 mm normally for RCC).
- All-in aggregate = fine + coarse together.

### ❖Required properties

- Hard, strong, durable, clean (free of clay, silt, organic matter, salts), angular & rough-textured (good bond), well graded (smallest voids), specific gravity 2.6–2.8, low water absorption (< 2%).
- Tests: sieve analysis (fineness modulus), crushing value, impact value, abrasion (Los Angeles), specific gravity, bulking (sand swells 20–40% at 5–8% moisture), flakiness/ elongation index.
- Well-graded aggregates → dense, strong, economical concrete (less cement).

### ❖5.3 Bricks

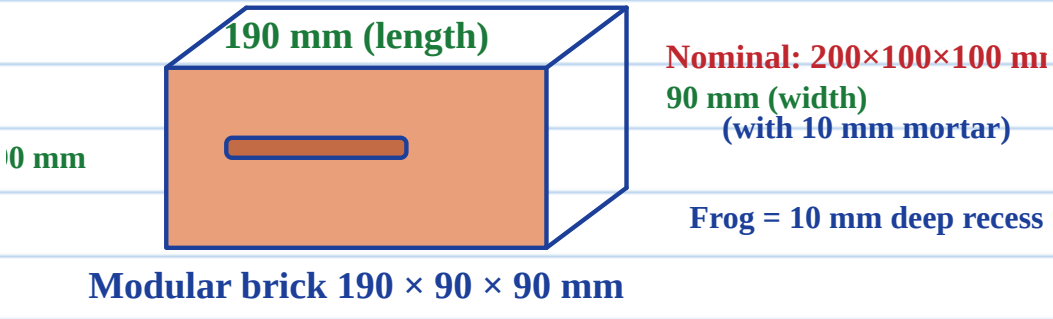



Fig 1.2 Standard (modular) brick

> **Brick** = a rectangular building unit made by moulding, drying and burning clay. Most popular wall material.

- **Standard (modular) size:** 190 × 90 × 90 mm (nominal 200 × 100 × 100 mm with mortar). Traditional brick 230 × 110 × 70 mm. **Weight ≈ 3 kg.**
- **Composition of good brick earth:** silica 50–60% (prevents shrinking), alumina 20–30% (plasticity), lime < 5%, iron oxide 5–6% (red colour, fusion), magnesia < 1%.
- **Manufacture:** (1) **preparation of clay** – digging, weathering, tempering; (2) **moulding** (hand/ machine); (3) **drying** (3–10 days); (4) **burning** in clamp/ Hoffmann kiln at 900–1100 °C.

### ❖Classification (IS 1077)

- **First class** – table-moulded, uniform shape, deep red, compressive strength > 10.5 $N/mm^2$ (class 10+), absorption ≤ 20% – for superstructure.
- **Second class** – ground-moulded, slightly distorted, absorption up to 22%; for plastered walls.
- **Third class** – under-burnt, soft, absorption high (25%) – temporary works.
- **Fourth class / jhama** – over-burnt, vitrified; crushed to coarse aggregate.
- **Others:** fly-ash bricks, hollow/ perforated, fire bricks, AAC blocks, concrete blocks, sand-lime bricks.

### ❖Qualities of good brick

- Uniform size & shape, sharp edges, **metallic ringing sound** when struck, no cracks, scratch-resistant, uniform colour, **absorbs ≤ 20% water in 24 h**, no efflorescence (white salt), compressive strength ≥ 3.5 – 35 $N/mm^2$.

### ❖5.4 Cement Concrete

> **Concrete** = Cement + Fine aggregate + Coarse aggregate + Water (+ admixtures). **Plain cement concrete (PCC)** is strong in compression but weak in tension; **RCC** = concrete + steel bars.

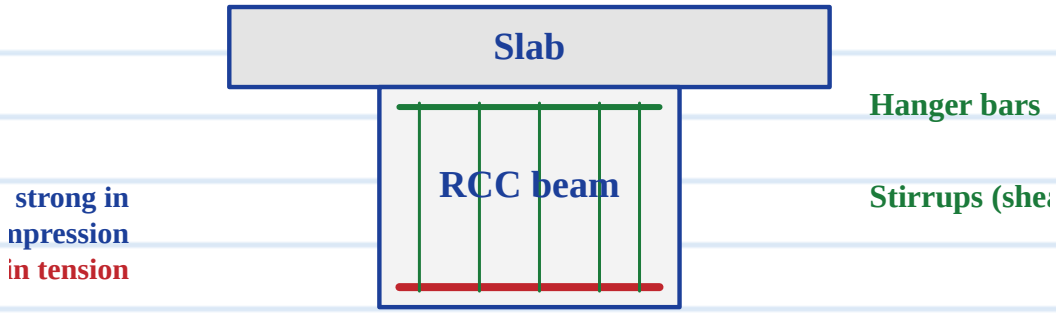



*Fig 1.3 Reinforced cement concrete (RCC) beam section*

## ❖Grades of concrete (IS 456)

- M20 (1 : 1.5 : 3) – "M" = mix; 20 = characteristic compressive strength ($N/mm^2$) at 28 days on 150 mm cubes. Other grades: M10, M15, M20, M25, M30 … M80.

| Grade | Ratio (C:S:A) | Use |
|---|---|---|
| M5 | 1 : 5 : 10 | Leveling/ PCC under foundation |
| M10 | 1 : 3 : 6 | PCC base |
| M15 | 1 : 2 : 4 | Plain concrete |
| M20 | 1 : 1.5 : 3 | RCC – general |
| M25 | 1 : 1 : 2 | RCC – heavy duty |

## ❖Properties

- **Workability** (slump test: 25–100 mm; compacting factor), **w/c ratio** (0.4–0.6; lower ratio = higher strength, **Abram's law**), **strength** (compressive – cube test), **durability**, **impermeability**, **shrinkage**, **creep**, density 2400 $kg/m^3$, RCC 2500 $kg/m^3$.

## ❖Steps in concreting

- **Batching** (weigh/ volume) → **mixing** (machine ≥ 2 min) → **transporting** → **placing** → **compaction** (needle vibrator – removes air voids) → **finishing** → **curing** (keep moist 7–28 days; strength gain, avoids cracks).
- **Cement : water** – 1 bag (50 kg) cement needs ≈ 25–30 ℓ water.
- **Admixtures:** plasticizers, retarders, accelerators, air-entraining agents; **mineral:** fly ash, silica fume.

## ❖Advantages & drawbacks

- Moldable, durable, fire-resistant, economical, high compressive strength; but heavy, weak in tension, needs curing, cracks with shrinkage.

### ❖S.S Steel

> Steel = alloy of iron + carbon (0.05–1.5%). Mild steel (0.15–0.25% C) is the most used in construction; strong in both tension and compression.

- Reinforcement bars (rebars): plain mild steel (Fe 250), TMT (Thermo-Mechanically Treated) bars – Fe 415, Fe 500, Fe 550; ribbed surface for bond; diameter 6–40 mm.
- Structural steel sections: I-section (ISMB), channel (ISMC), angle (ISA), T-section, flats, plates, hollow tubes; used in trusses, steel buildings, bridges, towers, industrial sheds.
- Properties: high strength, ductility, toughness, weldable, 100% recyclable; density 7850 $kg/m^3$, Young's modulus $2 \times 10^5$ $N/mm^2$, yield stress 250 $N/mm^2$ (Fe 250).
- Stress–strain: elastic limit → yield point → plastic → ultimate → rupture.
- Disadvantages: corrosion (needs paint/ galvanising), loses strength in fire (above 550 °C), costly.
- Corrosion protection: painting, galvanising, epoxy-coated bars, cathodic protection.
- Other materials (brief): timber (renewable, doors/ windows), glass, bitumen, tiles, aluminium, FRP / composites, geosynthetics.

## 6. Prefabricated Construction Techniques

> Prefabrication = manufacturing building components (walls, slabs, beams, columns, staircases, even whole rooms) in a factory or casting yard, transporting them to site and assembling/ erecting them.

### ❖Features of the technique

- Components are standardised, cast in moulds under controlled conditions; site work is only joining (dry/ wet joints, bolting, grouting).
- Types: precast concrete (RCC/ prestressed) panels, hollow-core slabs, precast walls, steel (pre-engineered) buildings, modular/ volumetric units (container-type), 3D printed walls.

### ❖Advantages

- Faster construction (time ↓ 30–50%), better quality control, less site labour, less wastage, less formwork & scaffolding, all-weather production, cost saving in repetition, cleaner site, eco-friendly.

### ❖Disadvantages

- High initial investment (plant, moulds), transport & lifting costs (cranes), joints need care (water leakage, structural continuity), limited design flexibility, skilled labour for erection, economical only for large volume / repetition.

### ❖Applications

- Mass housing, schools, hospitals, flyovers/ metro viaduct segments, precast piles, railway sleepers, compound walls, culverts, modular toilets, industrial sheds, PMAY (Light House Projects).

> Prefabrication → Factory made + Site assembled = Speed + Quality + Economy

## 7. Quick Revision & Important Questions

| Material | Key facts |
|---|---|
| Cement | Limestone+clay, kiln 1450 °C, IST ≥ 30 min, FST ≤ 10 h |
| Brick | 190×90×90 mm; absorption ≤ 20 %; > 3.5 $N/mm^2$ |
| Concrete | C+S+A+W; M20 = 1:1.5:3; cure 28 d |
| Steel | Fe 500 TMT; density 7850 $kg/m^3$ |

- Define civil engineering; explain its role in society.
- Write about the various disciplines of civil engineering and scope of each.
- List the components of a building; explain building planning principles.
- Write notes on cement, aggregates, bricks, concrete and steel.
- What is prefabrication? State its advantages and disadvantages.

## 8. Additional Topics for Better Marks

### ❖Mortar

- Mortar = binder (cement/ lime) + sand + water – used to join bricks/ stones and for plaster. Cement mortar 1:4 for brickwork; 1:6 for plaster; 1:3 for waterproof work.

### ❖Bonds in brick masonry

- **Stretcher** (long side seen) – 1/2 brick wall; **Header** (short side seen) – full-brick wall; **English bond** (alternate stretcher & header courses – strongest); **Flemish bond** (alternate header-stretcher in each course – good appearance).
- Rules: no continuous vertical joints, use **closers/ queen closers**, joints 10 mm, wet bricks before laying.

### ❖Timber, glass and others (brief)

- **Timber (teak, sal, deodar):** doors, windows, roofs; seasoning removes moisture; weak against fire/ termites. **Plywood, particle board** used in interiors.
- **Glass:** transparent; used for windows, facades; toughened/ laminated/ tinted types.
- **Bitumen:** black binder for roads & waterproofing. **Paints/ distemper** protect surfaces. **Tiles** for floors. **Aluminium** and **PVC** for doors/ windows.

### ❖Foundations – quick facts

- **Safe bearing capacity (SBC):** soft clay 50–100 $kN/m^2$; sand 200–400; hard rock > 3000 $kN/m^2$.
- Depth of foundation ≥ 0.9–1.2 m below ground; width of footing = Load / SBC.

## 9. Solved Problems

### ❖Problem 1 – Number of bricks

> Wall 10 m × 3 m × 0.2 m (modular brick with mortar 0.2×0.1×0.1 m = 0.002 $m^3$). Volume = 6 $m^3$ → bricks = 6/0.002 = **3000 bricks** (≈ 5% wastage → 3150).

### ❖Problem 2 – Materials for 1 $m^3$ of M20 concrete (1 : 1.5 : 3)

> Dry volume = 1.54 $m^3$. Sum of ratio = 5.5. Cement = 1.54 × 1/5.5 = 0.28 $m^3$ = 0.28 × 1440 = **403 kg ≈ 8 bags**; Sand = 1.54 × 1.5/5.5 = **0.42 $m^3$**; Coarse aggregate = 1.54 × 3/5.5 = **0.84 $m^3$**.

### ❖Problem 3 – Water for concrete

> Cement 400 kg, w/c = 0.5 → water = **200 litres**.

### ❖Problem 4 – Plinth area

> Plot 15 m × 20 m = 300 $m^2$; built-up per floor = 150 $m^2$; 3 floors = 450 $m^2$ → FSI = 450/300 = 1.5.

### ❖Problem 5 – Footing width

> Column load 600 kN; SBC = 150 $kN/m^2$ → area = 600/150 = 4 $m^2$ → square footing **2 m × 2 m**.

## 10. Sketch & Answer-Writing Tips

- Start every answer with a **definition**, give 4–5 clear points, end with **uses/advantages**.
- Draw neat diagrams: building section (label foundation, plinth, DPC, wall, lintel, slab), brick, RCC beam (steel at bottom).
- Mention **IS code numbers** and grades where possible (IS 456, IS 269, IS 1077, IS 1786).